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Java Conditional statement

Flow of control

Flow of control

- Selection (given choices)
  - if
  - if-else
  - switch.
- Iteration
- Jumps

1) If (condition) — True / False

```
{
    //statement
}
```

2) If-else — multiple choi

```
if (expression)
{
    statement
}
else
{
}
```

3. Nested if — first outer, then inner

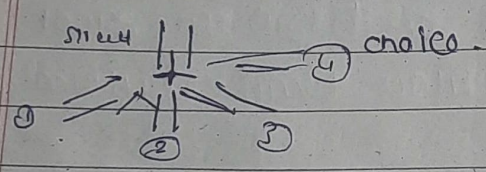
```
if (exp)
{
    if (exp)
    {
    }
}
else {
}
```

4. if-else-if ladder

```
if (i==5) {
    s.o.pln("i is 5")
}
else if (i==6) {
    s.o.pln()
}
else if (i==7) {
}
else {
}
```

until cond true, checks all if-else-if condn
when condn true, then other condn case
not checked.

5 Switch statement
- same as if-else-if ladder
- more optimized because of jump table.



Syntax:

```
switch (condition)
{
    case 1 :
        // statement
        break ;
    case 2 :
    :
    default :
        break;
}
```

```
int i = 8 ;
switch (i)
```

if no condn true, then default executed.
switch is optimized more than if-else ladder

- switch → Limited use case → should evaluate to byte, short, int, char, enumeration.
- No duplicate case allowed
- JDK 7 After — strings can also be used as switch expression
- After JDK-14 switch is enhanced too much.

* switch v/s if-else-if ladder

↓ equality & evaluate boolean exp also

can test equality

i = 4 — i > 4 ✗

switch(i)

if (i > 4) ✓

② switch is more efficient than if-else-if ladder

* switch provide direct access to expression instead of ~~checking~~ comparing each condn → done with JUMP TABLE

↳ JUMP table are always not efficient

case 1 → 1

case 2 → 2

JUMP table (for switch case faster)

Table-switch	Lookup-switch
- used when case values are continous (dense) (0, 1, 2, 3)	- used when cases values are not continuous (sparse values (like 1, 100, 10000)
- A table (array) stores jump address	- stores pairs (value → add)
- Directly jump to correct case using index	- Program search for matching value and then jumps
- Faster	- slower due to searching

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Nested switch

```
int j=5; int k=1;
switch (j) {
 case 4 :
        s.o.p ("j is 4");
        break;
 case 5 : switch(k) {
        s.o.p ("j is 5");
 default :
        s.o.p (j is not
```

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Iterations Statements
↳ loops

loops are used to repeat a set of instructions multiple times until a condition is satisfied.

Types of loops
↳ 1) While loop
↳ 2) Do-while loop
↳ 3) For-loop

1) while loop
Runs while condn is true
syntax:

```
while (Expression)
{
    // do something;
}
```